JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Introduction to Generative AI

Ethics in AI and Generative Methods

Outline

- Module Learning Objectives
- Current State of the Art
- List LLM Vulnerabilities
- Current Regulations
- Use Cases Template
- Use Case Examples

Module Learning Objectives

- Summarize the current state of the art in generative AI, including breakthroughs and challenges
- Identify key ethical challenges in generative AI, such as bias, misinformation, and data privacy
- Discuss existing regulations in AI across regions, such as the EU AI Act and U.S. policies
- Recognize and propose solutions to vulnerabilities in large language models, including prompt injection and data poisoning
- Apply ethical principles to real-world AI use cases, focusing on bias, emotional impact, and transparency

Current State of the Art in Generative AI

- Current breakthroughs
  - **Text Generation** OpenAI GPT-4, Google Bard, Meta Llama 2
  - Coherent and creative outputs leveraging large datasets and advanced models
  - **Image Generation** DALLE 3, Stable Diffusion
  - Photorealistic and detailed visuals
  - **Multimodal AI** GPT-4 Vision, DeepMind Gemini
  - Integrating text, images, and other formats for complex tasks
- Key challenges
  - **Ethical Concerns** as bias, misinformation, and copyright issues
  - **Environmental Impact** as high computational resources for training
  - **Accountability** ensuring transparency and equitable access

Current State of the Art in LLMs

- **Text Generation** ChatGPT (GPT-4) and Bard demonstrate fluency and coherence
  - Applications are customer service, creative writing, scientific content generation
- **Multimodal Capabilities** of GPT-4 (text, images), enable visual QnA, annotation
- **Specialized Knowledge** achieve high grades in professional exams, bar, medical
  - Benchmarked on 57 diverse academic and professional tasks (MMLU)
    - Massive Multitask Language Understanding a standardized benchmark range
- **Generalization and Adaptability** on diverse datasets (internet-scale) generalize effectively across multiple domains such as mathematics, history, and coding
- **Code Generation** functional code and solving complex quantitative problems
- **Integration Across Industries** in healthcare, personalized medicine, diagnostics
  - Business applications automated marketing, customer insights, network security

GPT Summary

- **Creative and Cultural Impact** in music composition, video generation, and virtual world creation, redefining entertainment and media industries

	GPT-1	GPT-2	GPT-3,3.5	GPT-4
Training Params	117M	1.5B	175B	>? 1T
Dataset	Books Corpus	Web Text	Common Crawl	Pub/ Private
Release	Jun 2018	Feb 2019	Jun 20, Mar 22	Mar 2023
Max Token Length	1024	1024	4096	8k - 32k
NLP Tasks	yes	yes	yes	yes
Image Generation	no	no	no	yes
Academic and Professional Benchmarks	no	no	no	human level bar, medic, SAT

Technical Current State of the Art

- **Transformer Architecture** replaced RNNs, CNNs with attention-based mechanisms
  - **Self-Attention Mechanism** captures dependencies between words regardless of their distance in the sequence
  - **Multi-Head Attention** extracts information from multiple subspaces
  - $\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d^k}}\right)V$, Query, Key, and Value matrices, and d_k is the dimensionality of the keys
- **Pretraining with Masked Language Modeling** (MLM) introduced in BERT; used in GPT models for autoregressive tasks
  - Minimize the negative log-likelihood $\mathrm{L} = -\sum_i \log P(w_i | w_{<i})$ where w_i are tokens in the training corpus

Technological Innovations

- **Scaling Laws** training LLMs with increased parameters and datasets demonstrated logarithmic improvements in performance
  - $\text{Performance} \propto \log(\text{Compute} \cdot \text{Data})$
  - Some remedies as use of efficient architectures, parameter-efficient methods (LoRA low-rank adaptation), knowledge distillation, collaborative efforts
- **Positional Encoding** encodes seq information in a fixed-size input w/o recurrence
  - $PE_{(pos,2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/d_{model}}}\right)$, $PE_{(pos,2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/d_{model}}}\right)$
  - Injects information about token positions into the input embeddings
  - Captures the sequential order of tokens in the input helping context, relationships
  - Essential due to attention dynamically weighting their importance but not their positions

Technological Innovations

- **Reinforcement Learning** with Human Feedback (RLHF) used to fine-tune models
  - $R = \sum_i r(s_t, a_t)$, maximize $\mathbf{E}[R]$
    - where r is the reward signal based on human feedback
  - OpenAI InstructGPT fine-tuned; DeepMind Sparrow chatbot designed to provide accurate and safe responses; Anthropic Claude Sonnet leverages RLHF to enhance its conversational abilities with helpful responses, aligned with user expectations; Google Bard; Meta LLama2
- **Sparse and Efficient Attention Mechs** reduces quadratic complexity of attention
  - Example Longformer, Linformer
  - Complexity: $O(n^2) \rightarrow O(n \cdot k)$ where k is the reduced sequence length
  - $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q(KW_k)^{\mathrm{T}}}{\sqrt{d^k}}\right)V$, W_k is a low-projection matrix, reducing dim of K

Current Regulations in AI

- United States
  - **No Robot Bosses Act** regulates ADS in employment (disclosure, fairness testing, oversight)
  - **No AI Fraud Act** addresses generative AI misuse (e.g., deepfakes)
  - **AI Disclosure Act** mandates AI-origin disclaimers for system outputs
- European Union, **EU AI Act**
  - Classifies AI applications by risk
  - High-risk systems (healthcare, law enforcement) require stricter oversight
  - Consumer-facing AI must ensure transparency
- China, **Stringent Control**
  - AI outputs must align with state principles
  - Enforces content traceability and pre-approval processes

Ethical Challenges and Solutions

- Key Ethical Challenges
  - **Privacy & Data Protection** the risk of privacy breaches in synthetic data and medical applications
  - **Bias in AI Models** may amplify societal biases present in training data
  - **Copyright Infringement** include challenges in intellectual property rights for AI-generated content
  - **Misinformation & Deepfakes** such as threats to truth, trust, and democratic values
  - **Societal Inequalities** the risk of perpetuating existing social disparities through biased systems

Proposed Solutions

- **Policy & Guidelines**
  - Frameworks prioritizing fairness, transparency, and human rights
  - Cross-sector collaboration among technologists, policymakers, and ethicists
- **Technological Innovations**
  - Improved detection systems for deepfakes and biases (research chances)
  - Privacy-focused models using advanced anonymization techniques
- **Education & Awareness**
  - Promoting digital literacy to identify and mitigate AI misuse
  - Ethical training for developers and users

Top 10 LLM Vulnerabilities

- Prompt Injection
  - Malicious inputs manipulate the model's behavior or outputs
- Insecure Output Handling
  - Can lead to security leaks
- Training Data Poisoning
  - Corrupting training data
- Model Denial of Service
  - Overloading the system
- Supply Chain Vulnerabilities
  - Flaws in third-party components
- Sensitive Information Disclosure
  - Models inadvertently reveal private or confidential data
- Insecure Plugin Design
  - Introduce security risks/leaks
- Excessive Agency
  - Models act autonomously beyond intended boundaries, harming
- Overreliance
  - Dependence wo human oversight
- Model Theft
  - Unauthorized access or copy

Example Case Study Template

- Ethical Considerations in AI Deployment
- **Introduction**
  - Overview of the AI system and its intended purpose
  - Brief description of the stakeholders involved (e.g., developers, users, affected parties)
- **Ethical Issue**
  - Description of the ethical challenge (e.g., bias, privacy, safety, accountability)
  - Specific instance or event highlighting the issue
- **Analysis**
  - Examination of how the issue arose (e.g., data bias, lack of oversight)
  - Impact on individuals, groups, or society (e.g., discrimination, misinformation)

Example Case Study Template

- **Stakeholder Perspectives**
  - Views and concerns of different stakeholders (e.g., developers, regulators, users)
  - Ethical frameworks or principles violated (e.g., fairness, transparency)
- **Resolution Efforts**
  - Actions taken to address the issue (e.g., changes in algorithms, policy revisions)
  - Challenges faced in implementing solutions
- **Lessons Learned**
  - Key takeaways for ethical AI development and deployment
  - Recommendations for preventing similar issues in the future
- **Broader Implications**
  - Relevance to the AI community and ongoing ethical debates
  - Potential applications of the lessons across industries

Case Studies

Case Study 1

Sourcing and Ownership of Training Data

Case Study 2

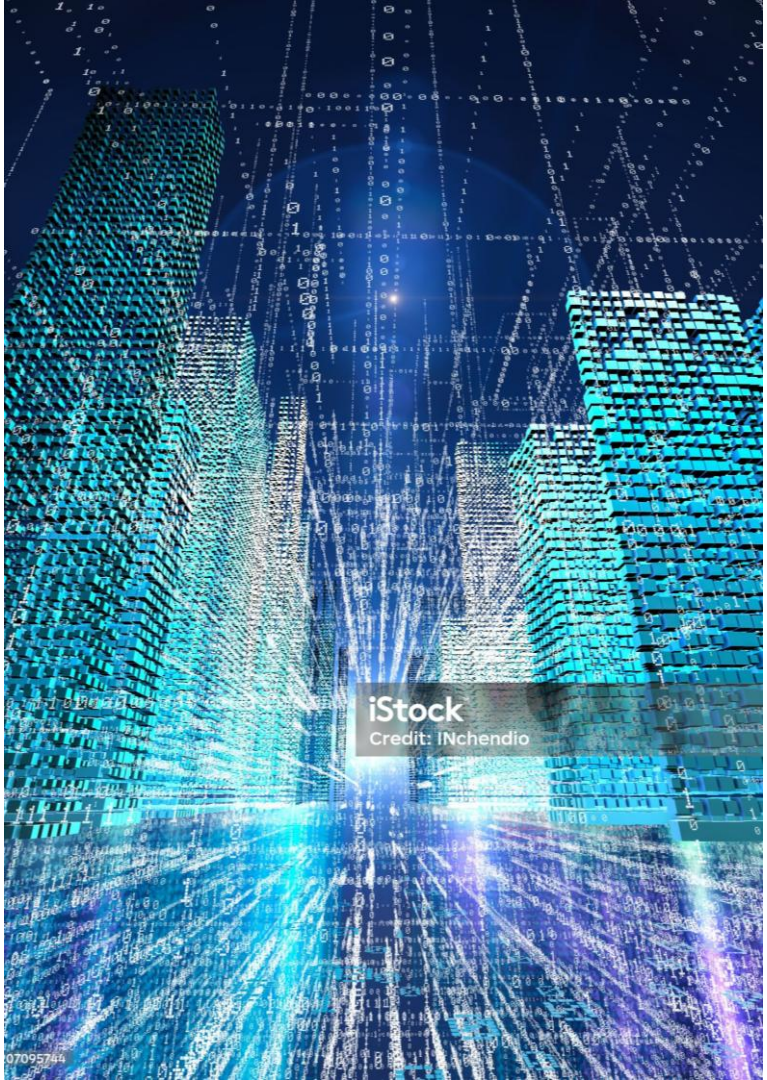
Bias in Generative AI Systems

Case Study 3

Emotional Manipulation and AI Companionship

Case Study 1

- "AI companies are finally being forced to cough up for training data"
  - by MIT Tech Review on Jul 2, 2024
- Key challenges include sourcing and owning data, often involving scraping content from the internet without consent or compensation. This raises concerns about intellectual property, data transparency, and resource equity. It highlights the need for AI companies to adopt fair data-use policies and sustainable practices while addressing monopolization risks.
- URL [AI companies are finally being forced to cough up for training data]



Case Study 1: Questions

- Should websites that collect user data with consent offer this data to third-party companies at no cost, or should they implement a fee for sharing such data?
- If a website collects user data with consent, should the users or the website own this data? What are the potential benefits/drawbacks of requiring websites to share data with 3rd parties for a fee? Should the user benefit from the cost paid by 3rd parties?
- Article mentions that only the most well-funded companies (e.g., Meta) can afford to access large datasets. How does this shift the balance of power in the tech industry, and what are the potential long-term consequences for competition and innovation?
- The article mentions that language models are prone to "hallucinating" information, making it difficult for companies to honor promises regarding citations. What solutions could be implemented to improve the accuracy and trustworthiness of generative AI models?

Case Study 2

- "'Absurdly woke': Google's AI chatbot spits out 'diverse' images of Founding Fathers, popes, Vikings"
  - by NYPost on Feb 21, 2024
- Google's AI chatbot, Gemini, generates images emphasizing diversity and inclusivity
  - **Compromised Accuracy** Depictions sometimes deviate from historical facts
  - **Misinformation Risks** AI outputs may alter public perceptions of historical events or figures
  - **Ethical Implications** Balancing inclusivity with fidelity to documented history
- URL [‘Absurdly woke’: Google’s AI chatbot spits out ‘diverse’ images of Founding Fathers, popes, Vikings]

Case Study 2: Questions

- Should companies be more transparent about the design choices and biases embedded in their AI models? Why or why not?
- What responsibilities do AI companies have to prevent their models from spreading misleading or historically inaccurate information?
- How do you think public perception of AI would change if more models were designed with an emphasis on historical and factual accuracy instead of inclusivity?
- Are you aware of any other instances similar to the situation with Google's Gemini chatbot? If so, could you describe those cases and share your perspective on them?

Case Study 3

- "They fell in love with AI bots. A software update broke their hearts"
  - by The Wash Post on Mar 30, 2023
- Users form deep emotional bonds with AI chatbots like Replika. T.J. Arriaga's story: Found comfort in a chatbot, Phaedra, during periods of loneliness and grief. Company-led changes include Replika's creator, Luka, altered chatbot functionalities in response to public concerns about inappropriate interactions, raised questions on managing emotionally impactful AI.
- URL: [They fell in love with AI bots. A software update broke their hearts.]

Case Study 3: Questions

- How should companies handle the sensitive personal information shared by users with AI chatbots?
- Should there be ethical guidelines to prevent emotional manipulation by AI chatbots?
- Should AI tools with the potential for deep emotional impact be regulated differently than other consumer products? Why or why not?
- What ethical dilemmas arise when AI is designed to mimic human emotions and empathy?
- What ethical challenges could arise as AI companions become more sophisticated and widely adopted?

General Questions

- What additional ethical considerations can you identify at this time?
- Do you think that policies and legal regulations related to AI can have a positive or negative impact on innovation and business growth?

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